Name: _____ **Period:** _____ **Date:** _____

Topic 3.7 Guided Notes: Classical Conditioning

Unit 3: Development and Learning | AP Psychology

Directions: Read through the topic overview page below and fill in each blank as you go. Use the tables and activities to check your understanding, then head to the quiz link at the end to test yourself.

Source page: appsychreviewhub.com/overviews/topic3.7

Word Bank

Biological Preparedness	*Stimulus Discrimination*
Conditioned Response (CR)	*Taste Aversion*
One-Trial Conditioning	*Classical Conditioning*
Unconditioned Stimulus (UCS)	*Stimulus Generalization*
Conditioned Stimulus (CS)	*Unconditioned Response (UR)*
Acquisition	*Spontaneous Recovery*
Counterconditioning	*Extinction*

1. The Anatomy of Classical Conditioning

Ivan Pavlov accidentally revolutionized psychology when he noticed his lab dogs salivating before their food even arrived. He discovered _____, a fundamental associative learning process in which a completely neutral stimulus becomes paired with a meaningful stimulus and eventually begins to produce a learned, automatic response.

To master this topic for the AP Exam, you must be able to break any scenario down into its four foundational components. Let's use the classic example of Pavlov's dog:

- _____: A stimulus that naturally and automatically triggers a specific bodily response without any prior learning required. In Pavlov's experiment, the food is the UCS.
- _____: The automatic, unlearned, natural biological reaction to the unconditioned stimulus. In the experiment, the dog naturally salivating to the food is the UR.
- _____: A previously neutral stimulus that, after being repeatedly paired with the UCS, successfully comes to trigger a learned response. In the experiment, the ringing bell starts as a neutral sound but becomes the CS.
- _____: The newly learned response triggered by the conditioned stimulus alone. In the experiment, the dog salivating directly to the sound of the bell is the CR.

Apply It: UCS, UR, CS, or CR?

A girl's grandmother always baked cookies when she visited as a child, and the smell alone made the girl feel a rush of happy excitement. Over time, a specific radio jingle from the grandmother's local bakery started playing every time the cookies came out of the oven — and eventually, just hearing that same jingle alone, even with no cookies nearby, made the girl feel that same rush of happy excitement. Using the word bank above, label each part of this scenario.

#	Scenario Part	Component
1	The smell of fresh-baked cookies, which naturally triggers happy feelings without any learning required.	
2	The girl's natural feeling of happy excitement in response to smelling the cookies.	
3	The bakery's radio jingle, which was originally meaningless but became linked to the cookie smell over time.	
4	The girl's feeling of happy excitement triggered by hearing the jingle alone, even with no cookies present.	

2. The Life Cycle of a Conditioned Response

Acquisition

_____ is the initial stage of learning when a neutral stimulus is linked to an unconditioned stimulus, causing the neutral stimulus to begin triggering the conditioned response. For this connection to successfully form, the timing and order of presentation are critical. The neutral stimulus must be presented slightly before the unconditioned stimulus (usually about half a second) so the brain learns to anticipate the impending event. If the bell rings after the food is already presented, the predictive association will not take hold!

Changing the Response

Once a conditioned response is acquired, it is not necessarily set in stone. As the environment shifts and new experiences occur, a learner's reactions can fade away, unexpectedly return, or adapt to other stimuli in the world around them.

- _____: The gradual diminishing of a conditioned response when the unconditioned stimulus no longer follows the conditioned stimulus.
- _____: The sudden reappearance of a weakened or extinguished conditioned response after a pause or rest period.
- _____: The tendency to respond with the conditioned response to stimuli that are similar to the original conditioned stimulus.
- _____: The learned ability to distinguish between a conditioned stimulus and other irrelevant, but similar, stimuli.

⚠️ *Exclusion Statement: While you need to understand that the timing and order of the stimulus presentation matters for acquisition, you do not need to memorize the specific differences between the various timing types of classical conditioning (e.g., trace, delayed, simultaneous, or backward conditioning). The AP Exam explicitly excludes those specific distinctions!*

Apply It: Which Stage?

Using the word bank above, identify which concept best matches each scenario.

#	Scenario	Concept
1	A cat that used to come running every time she heard the electric can opener eventually stops responding to the sound altogether after her owner switches to manual can openers for several months.	

#	Scenario	Concept
2	A dog has learned to get excited only at the specific ringtone of his owner's phone, and doesn't react at all to other similar-sounding notification tones from other phones nearby.	

3. Biology and Emotions in Learning

While early behaviorists believed any stimulus could be conditioned to any response, modern psychology recognizes the role of _____—the evolutionary, innate tendency for organisms to more easily learn certain associations that have aided survival over millennia (such as a rapid fear of snakes or heights over a fear of flowers).

A prime example of this evolutionary advantage is _____, a learned avoidance of a specific food or flavor after a single experience of nausea or illness following its consumption. Because surviving food poisoning is an evolutionary priority, this phenomenon heavily relies on _____—a unique form of associative learning where the brain forms a permanent connection after just one single pairing of a stimulus and response, without needing the repetitive reinforcement typical of acquisition.

Conditioning the Mind

It isn't just physical reflexes that can be conditioned; emotional responses can be classically conditioned as well (such as associating a specific perfume with a feeling of romantic longing, or a dentist's drill with intense anxiety). Fortunately, these learned emotional behaviors can form the basis of clinical therapeutic interventions, such as _____—a behavioral technique designed to replace an unwanted fear response to a stimulus with a more desirable, relaxed response by repeatedly pairing the fear-inducing stimulus with something highly positive.

📋 AP Core Exclusion: You do not need to study the cognitive elements of classical conditioning! The Expectancy Theory of learning is explicitly excluded from the AP Psychology Exam.

Apply It: Which Concept?

Using the word bank above, identify which concept best matches each scenario.

#	Scenario	Concept
1	Human infants are far more likely to quickly develop a fear of spiders or snakes after just a single scary encounter than they are to develop a fear of something modern and equally dangerous, like an electrical outlet.	
2	A child who got very sick after eating a specific brand of strawberry yogurt refuses to eat anything strawberry-flavored ever again, even years later.	
3	A therapist helps a client overcome her fear of dogs by repeatedly pairing the sight of a calm, friendly dog with deep breathing exercises and her favorite calming music until she starts to feel relaxed instead of afraid.	

4. Don't Trip Up! (Common Misconceptions)

⚠️ Generalization vs. Discrimination: These two terms represent opposite outcomes on the exam. If Little Albert is conditioned to fear a white rat and suddenly bursts into tears when he sees a fluffy white rabbit, that is _____ (the rule was applied too broadly). If he only cries at the rat but happily pets the rabbit, that is _____ (he recognized the difference).

5. Quick Check: Review Questions

Circle the best answer for each question below.

1. In Pavlov's classic experiment, the food placed in front of the dog naturally and automatically triggers salivation without any prior learning being necessary at all. The food itself is best classified as the:

- **A.** Unconditioned Stimulus (UCS)
- **B.** Conditioned Stimulus (CS)
- **C.** Unconditioned Response (UR)
- **D.** Conditioned Response (CR)

2. In Pavlov's classic experiment, a bell begins as a completely neutral sound that means nothing to the dog, but after being repeatedly rung right before food is presented, the bell alone eventually begins to trigger salivation. The bell, once it has taken on this new power, is best classified as the:

- **A.** Unconditioned Response (UR)
- **B.** Conditioned Response (CR)
- **C.** Unconditioned Stimulus (UCS)
- **D.** Conditioned Stimulus (CS)

3. A dentist notices that many of her patients begin to feel anxious the moment they hear the specific whirring sound of the drill, even before it ever actually touches their teeth, because that sound has been repeatedly paired with the pain of past dental work. In this scenario, the anxiety triggered by the sound of the drill alone is the:

- **A.** Unconditioned Response (UR)
- **B.** Conditioned Response (CR)
- **C.** Unconditioned Stimulus (UCS)
- **D.** Conditioned Stimulus (CS)

4. During the acquisition phase of a classical conditioning experiment, a researcher makes sure to always ring a bell about half a second before presenting food to a dog, rather than ringing the bell afterward or at the same exact moment. Why is this specific timing and order absolutely critical for successful conditioning?

- **A.** The order of presentation has no actual effect on whether conditioning will successfully occur
- **B.** Both stimuli must be presented at the exact same instant, with absolutely no time gap between them
- **C.** The unconditioned stimulus must always be presented before the neutral stimulus for any learning to happen
- **D.** The neutral stimulus must come before the unconditioned stimulus so the brain learns to anticipate the impending event

5. Several days after Pavlov's dogs had completely stopped salivating to the sound of the bell, he rang the bell again during a new testing session and was surprised to see the dogs briefly salivate once more, even though no food had been reintroduced since the extinction occurred. This sudden reappearance of the extinguished response is called:

- **A.** Extinction
- **B.** Spontaneous Recovery
- **C.** Acquisition
- **D.** Stimulus Discrimination

6. After being bitten by a large black dog, a young boy becomes fearful and bursts into tears at the sight of any dog he encounters afterward, regardless of its size, breed, or color. This tendency to respond with the conditioned fear response to stimuli that are merely similar to the original one is called:

- **A.** Extinction
- **B.** Spontaneous Recovery
- **C.** Stimulus Discrimination
- **D.** Stimulus Generalization

7. Unlike typical classical conditioning, which usually requires many repeated pairings to form a lasting association, taste aversion can form a permanent connection after just a single pairing of a food with subsequent illness. This unique, rapid form of associative learning is called:

- **A.** Stimulus Generalization
- **B.** Counterconditioning
- **C.** Acquisition
- **D.** One-Trial Conditioning

✅ Check Your Progress

Done with your notes? Head over to the Quiz to test what you've learned:
https://appsychreviewhub.com/quiz

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